

DOOB'S PROOF OF MLE CONSISTENCY

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ABSTRACT. A concise presentation of Doob's proof of the consistency of Fisher's maximum likelihood estimator.

1. INTRODUCTION

The consistency of Fisher's maximum likelihood estimator (MLE) [10] is a basic result discovered more than a century ago. This method is typically treated as a special case of M -estimators, and derived via standard techniques dating back to Cramér [6] and Wald [24]. A modern treatment is van der Vaart [23]. Because of its intuitive appeal and broad applicability, the method is a dominant tool of statistical inference.

In teaching this material, I found Doob's consistency proof [9] of MLE convergence more motivated and accessible to students, particularly for students already familiar with information and entropy as part of a data science or machine learning course. The fact that the proof covers general parameter θ spaces and general state X spaces with no extra effort only adds to its appeal.

The aim of this note is to present this proof, together with some historical terminology remarks, in a self-contained manner. By using the law of large numbers in its strong form, we derive MLE consistency in its strong form.

We start with the setup and assumptions, then state and prove consistency. After that, we give a historical account of entropy terminology, which is an interesting story in itself, then we end by showing the assumptions hold for the multivariate normal case.

2. SURPRISAL AND INFORMATION

A random variable X is sampled repeatedly, resulting in independent and identically distributed copies $X_1, X_2, \dots$ of X . We assume the density of X is $p(X, \theta^*)$, one of a parametric family $p(X, \theta)$ of densities, and we wish to recover θ^* from the samples $X_1, X_2, \dots$.

Stated in this manner, all we have to work with are two items: the true expectation E_* and the family $p(X, \theta)$. While $p(X, \theta)$ may be chosen arbitrarily, based on prior knowledge, we have no control over or knowledge of E_* . Given this, how do we use the samples $X_1, X_2, \dots$ to recover θ^* ?

Define the *surprisal* [18, 21]

$$S(X, \theta) = \log(1/p(X, \theta)).$$

This terminology is intuitive: low probability equals high surprisal. Then it is natural to characterize θ^* as a minimizer of the expected surprisal $E_*(S(X, \theta))$. More precisely, we say θ^* is the *true parameter* if $S(X, \theta)$ is integrable for all θ and θ^* is the *unique* minimizer of $E_*(S(X, \theta))$. In particular, if θ^* is the true parameter and $\theta \neq \theta^*$, then $p(X, \theta) \neq p(X, \theta^*)$.

Equivalently, given θ^* , the *relative information* is

$$(1) \quad I(\theta^*, \theta) = E_*(S(X, \theta)) - E_*(S(X, \theta^*)).$$

The relative information is the excess expected surprisal of θ over θ^* . Then θ^* is the true parameter iff $I(\theta^*, \theta) \geq 0$ for all θ , with $I(\theta^*, \theta) = 0$ only if $\theta = \theta^*$. The relative information I appears in many fields, in different forms. Because of widely differing terminology, below we give an account of its history.

A special case is when there is¹ a background expectation E such that

$$E_\theta(f(X)) = E(p(X, \theta)f(X))$$

¹While $E_\theta(1) = 1$, E need not satisfy $E(1) = 1$, $E(1) = \infty$ is allowed.

is an expectation for all θ . When this holds and $E_{\theta^*} = E_*$, the relative information takes the familiar form

$$I(\theta^*, \theta) = E \left(p(X, \theta^*) \log \left(\frac{p(X, \theta^*)}{p(X, \theta)} \right) \right),$$

$I(\theta^*, \theta) \geq 0$ for all θ , and $I(\theta^*, \theta) = 0$ implies² $p(X, \theta) = p(X, \theta^*)$ [7, 8]. When $\theta \neq \theta^*$ implies $p(X, \theta) \neq p(X, \theta^*)$, it follows θ^* is the true parameter.

The simplest example is as follows. A vector $\theta = (\theta_1, \theta_2, \dots, \theta_d)$ is a probability vector if its components are nonnegative and sum to one. Suppose $\theta^* = (\theta_1^*, \theta_2^*, \dots, \theta_d^*)$ is a probability vector and a random variable X satisfies $P_*(X = i) = \theta_i^*$, $i = 1, 2, \dots, d$. Then the mass function of X is $p(X, \theta^*) = \theta_X^*$. If θ is any other probability vector, the surprisal is $S(X, \theta) = \log(1/\theta_X)$, and

$$(2) \quad I(\theta^*, \theta) = \sum_{i=1}^d \theta_i^* \log(\theta_i^*/\theta_i).$$

Turning back to the general setting, the *likelihood* and *mean surprisal* are

$$L_n(\theta) = \prod_{k=1}^n p(X_k, \theta), \quad \bar{S}_n(\theta) = \frac{1}{n} \log(1/L_n(\theta)) = \frac{1}{n} \sum_{k=1}^n S(X_k, \theta).$$

The mean surprisal is also called the negative log-likelihood.

By definition, θ^* minimizes $E_*(S(X, \theta))$. By the law of large numbers, $\bar{S}_n(\theta)$ converges to $E_*(S(X, \theta))$ as $n \rightarrow \infty$, with probability one. Hence it is natural to estimate θ^* by a minimizer of $\bar{S}_n(\theta)$ or, equivalently, a maximizer of $L_n(\theta)$: For each $n \geq 1$, a *maximum likelihood estimator (MLE)*, if it exists, is a parameter $\hat{\theta}_n$, depending on the samples, maximizing $L_n(\theta)$ over all θ . Under **A** and **B** below, with probability one, an MLE sequence $\hat{\theta}_n$ exists for sufficiently large n .

3. CONSISTENCY

In addition to the existence of the true parameter, we make two assumptions. A function $F(\theta)$ is *proper* if its sublevel sets $F(\theta) \leq c$ are compact subsets of the parameter space, for every level c . Since $\hat{\theta}_n$ should minimize $\bar{S}_n(\theta)$, this leads us to the first assumption

A. (Properness): There is an $N \geq 1$, depending on the samples, and a proper $F(\theta)$, not depending on the samples, such that, with probability one,

$$(3) \quad F(\theta) \leq \bar{S}_n(\theta), \quad \text{for all } \theta \text{ and } n \geq N.$$

We will need the convergence of $\bar{S}_n(\theta)$ to $E_*(S(X, \theta))$ across θ , in the sense

$$(4) \quad \bar{S}_n(\theta) \rightarrow E_*(S(X, \theta)), \quad \text{as } n \rightarrow \infty, \quad \text{for all } \theta,$$

with probability one. For this we need the second assumption

B. (Continuity): The parameter space is separable and there is a continuous $f(\theta, \theta')$ and an integrable $g(X)$ satisfying

$$|S(X, \theta) - S(X, \theta')| \leq f(\theta, \theta')g(X) \quad \text{and} \quad f(\theta, \theta) = 0.$$

A separability argument [19] then establishes (4) with probability one.

Under **A** and **B**, an MLE $\hat{\theta}_n$ exists³ for $n \geq N$, with probability one. **A** and **B** are typically checkable for *exponential families* [5, 15, 19, 23], the densities $p(X, \theta)$ where $S(X, \theta)$ is a sum of products of the form $f(\theta)g(X)$.

Theorem (Consistency). *Let $X_1, X_2, \dots$ be independent and identically distributed, and let $p(X, \theta)$ be a parametric family with true parameter θ^* . Under **A** and **B**, with probability one, an MLE sequence $\hat{\theta}_n$ may be defined for sufficiently large n , and $\hat{\theta}_n$ converges to θ^* as $n \rightarrow \infty$.*

²Apply Jensen's inequality with the strictly convex $\log(1/p)$.

³A proper lower semicontinuous function has a minimizer.

Proof. By the law of large numbers,

$$(5) \quad \frac{1}{n} \sum_{k=1}^n g(X_k) \rightarrow E_*(g(X)), \quad \text{as } n \rightarrow \infty,$$

with probability one. Fix samples for which (3), (4) and (5) hold. We show $\hat{\theta}_n \rightarrow \theta^*$ for these samples. By (3),

$$F(\hat{\theta}_n) \leq \bar{S}_n(\hat{\theta}_n) \leq \bar{S}_n(\theta^*), \quad \text{for } n \geq N.$$

Since the right side is bounded as $n \rightarrow \infty$, and $F(\theta)$ is proper, the sequence $\hat{\theta}_n$ lies in a compact set in the parameter space.

Suppose θ is a limit point of $\hat{\theta}_n$. Then $\hat{\theta}_n \rightarrow \theta$ along a subsequence. By B,

$$(6) \quad \bar{S}_n(\theta) - \bar{S}_n(\theta^*) \leq \bar{S}_n(\theta) - \bar{S}_n(\hat{\theta}_n) \leq f(\theta, \hat{\theta}_n) \cdot \frac{1}{n} \sum_{k=1}^n g(X_k).$$

Since the left side of (5) converges to $E_*(g(X))$, and $f(\theta, \hat{\theta}_n) \rightarrow 0$ along a subsequence, the right side of (6) goes to zero along a subsequence. By (1) and (4), the left side of (6) converges to $I(\theta^*, \theta)$, hence $I(\theta^*, \theta) \leq 0$. But $I(\theta^*, \theta) \geq 0$, hence $I(\theta^*, \theta) = 0$, showing $\theta = \theta^*$. Thus $\hat{\theta}_n \rightarrow \theta^*$ as $n \rightarrow \infty$. $\square$

4. TERMINOLOGY HISTORY

The relative information $I(\theta^*, \theta)$ is often called “relative entropy” [5, 15]. The term “entropy”, originating in thermodynamics, was coined by Clausius in his 1865 paper [4]. Boltzmann, in his 1872 paper [2], derived his H theorem, using a quantity H proportional to the negative of entropy. Gibbs, in his 1902 book [11], building on Boltzmann’s 1877 combinatorial interpretation [3] and Planck’s 1901 paper [17], flipped Boltzmann’s sign convention, writing the entropy H as done today.

Shannon, in his 1948 paper [20], independently defined

$$H(p) = - \sum_{i=1}^d p_i \log p_i.$$

Von Neumann had earlier defined quantum entropy in his 1932 book [16], and when Shannon consulted him around 1939–1940 about terminology for H , von Neumann suggested “entropy”, which Shannon adopted [22]. However, in his remark “ H is then, for example, the H in Boltzmann’s H theorem”, Shannon conflates his H with Boltzmann’s H .

The relative information $I(\theta^*, \theta)$ is jointly convex in the densities $p(X, \theta^*)$ and $p(X, \theta)$, a fact that is apparent in the special case (2), whereas $H(p)$ above is strictly concave in p . Because of this, calling both quantities “entropy” creates a mismatch. Indeed, based on the curvature, one might argue it is more natural to call the *negative* of (2) the *relative entropy*. The mismatch is also in Python: `entropy(p)` returns the concave H , but `entropy(p,q)` returns the convex I .

This confusion can be dissolved by remembering entropy and information are opposites: entropy is concave and information is convex. More generally, this error is a common psychological conflation: When measuring the temperature, are we measuring how cold it is, or how hot it is? Of course we are doing both, but this doesn’t mean cold equals hot. In fact, cold equals minus hot.

While many authors’ terminology conflicts with ours, some authors align partially with our usage. Schervish, in his 1995 text [19], uses “Kullback-Leibler information” for I , which is more descriptive than the widely used but opaque “Kullback-Leibler divergence” [1]. Kullback himself, in his 1959 monograph [14], refers to I as an “information function”. Donsker-Varadhan, in their 1975 large deviations paper [8], denote their rate function by I rather than H ; the relative information is a special case of their I .

Regarding Doob’s 1934 proof, this was at the dawn of abstract probability theory, within the milieu of Daniell, Fréchet, Hopf, Khintchine, Kolmogorov, Lévy, and Wiener, and within a year of Kolmogorov’s 1933 foundational extension theorem [13].

5. MULTIVARIATE NORMAL

Let μ be in $\mathbf{R}^d$, Q a positive definite symmetric $d \times d$ matrix, and $\theta = (\mu, Q)$. Let $v \cdot w$ be the dot product in $\mathbf{R}^d$. With X in $\mathbf{R}^d$, the *normal density with mean μ and variance Q* is $p(X, \theta) = e^{-S(X, \theta)}$ where

$$(7) \quad S(X, \theta) = \frac{1}{2} \log \det(2\pi Q) + \frac{1}{2}(X - \mu) \cdot Q^{-1}(X - \mu).$$

This defines the parametric family.

A random variable X in $\mathbf{R}^d$ is *normally distributed with parameter $\theta^* = (\mu^*, Q^*)$* if its moment-generating function is

$$(8) \quad E_*(e^{v \cdot X}) = \exp \left(v \cdot \mu^* + \frac{1}{2} v \cdot Q^* v \right).$$

Suppose X is normally distributed with parameter $\theta^* = (\mu^*, Q^*)$ as in (8), and let $X_1, X_2, \dots$ be independent and identically distributed copies⁴ of X . We show θ^* is the true parameter, and we verify **A** and **B**.

Assumption **B** follows from (7). To establish **A**, we review basic facts [12], stated as Lemmas, with proofs in the Appendix.

Given vectors v, w in $\mathbf{R}^d$, the *tensor product* $v \otimes w$ is the matrix whose ij -th entry is $v_i w_j$. Then $v \otimes v$ is symmetric,

$$(9) \quad v \cdot Qv = \text{trace}(Q(v \otimes v)), \quad v \otimes w + w \otimes v \leq \epsilon v \otimes v + \frac{1}{\epsilon} w \otimes w,$$

and, by the eigenvalue decomposition, every symmetric matrix is a linear combination of matrices of the form $v \otimes v$.

Given symmetric matrices Q, R, S , we say $Q \geq 0$ if $v \cdot Qv \geq 0$ for all v , and $Q \geq R$ if $Q - R \geq 0$. If $Q \geq \delta I$ for some $\delta > 0$, we say $Q > 0$. If $Q \geq 0$ and $R \geq S$, then $\text{trace}(QR) \geq \text{trace}(QS)$ and $\text{trace}(QR) = \text{trace}(RQ)$.

The determinant $\det(Q)$ is the product of the eigenvalues of Q .

Lemma 1. *With Q invertible symmetric and t a scalar, $\det(Q + tv \otimes v) = \det(Q) (1 + tv \cdot Q^{-1}v)$.*

Lemma 2. *If $S > 0$ and $Z > 0$, then $\det(SZS) = \det(Z) \det(S)^2$.*

Lemma 3. *For c scalar, let $\epsilon = e^{-c}$ and $h(\lambda) = \log \lambda + 1/\lambda$, $\lambda > 0$. Then $h(\lambda)$ is minimized iff $\lambda = 1$, and $h(\lambda) \leq c$ implies $\epsilon \leq \lambda \leq 1/\epsilon$.*

Lemma 4. *For $Q^* > 0$ and c scalar, let $\epsilon = e^{-2c} \det(2\pi Q^*)$ and*

$$(10) \quad F_*(Q, Q^*) = \frac{1}{2} \log \det(2\pi Q) + \frac{1}{2} \text{trace}(Q^{-1}Q^*), \quad Q > 0.$$

Then $F_(Q, Q^*)$ is minimized iff $Q = Q^*$, and $F_*(Q, Q^*) \leq c$ implies $\epsilon Q^* \leq Q \leq Q^*/\epsilon$.*

Let $f(\mu) = Q^* + (\mu^* - \mu) \otimes (\mu^* - \mu)$. Given $\theta = (\mu, Q)$, let

$$F(\mu, Q) = F_*(Q, f(\mu)) = F_*(Q, Q^*) + \frac{1}{2}(\mu^* - \mu) \cdot Q^{-1}(\mu^* - \mu).$$

By Lemma 4, (μ^*, Q^*) is the unique minimizer of $F(\mu, Q)$.

In (8), multiply both sides by $e^{-v \cdot \mu}$, then replace v by tv and differentiate twice at $t = 0$. This yields

$$(11) \quad E_*(X) = \mu^*, \quad E_*((X - \mu) \otimes (X - \mu)) = f(\mu).$$

It follows $E_*(S(X, \theta)) = F(\mu, Q)$, and $\theta^* = (\mu^*, Q^*)$ is the true parameter.

If $F(\mu, Q) \leq c$, then $F_*(Q, Q^*) \leq c$, hence $\epsilon Q^* \leq Q \leq Q^*/\epsilon$, confining Q to a compact subset of the space of positive definite matrices. From this,

$$\begin{aligned} c &\geq F_*(Q, Q^*) + \frac{1}{2}(\mu^* - \mu) \cdot Q^{-1}(\mu^* - \mu) \\ &\geq F_*(Q^*, Q^*) + \frac{\epsilon}{2}(\mu^* - \mu) \cdot (Q^*)^{-1}(\mu^* - \mu), \end{aligned}$$

confining μ to a compact subset of $\mathbf{R}^d$. Thus $F(\mu, Q)$ is proper.

⁴This setup is a special case of the Kolmogorov extension theorem.

With

$$\hat{\mu}_n = \frac{1}{n} \sum_{k=1}^n X_k, \quad \hat{Q}_n = \frac{1}{n} \sum_{k=1}^n (X_k - \hat{\mu}_n) \otimes (X_k - \hat{\mu}_n),$$

let $\hat{\theta}_n = (\hat{\mu}_n, \hat{Q}_n)$ and

$$Q_n = \frac{1}{n} \sum_{k=1}^n (X_k - \mu) \otimes (X_k - \mu).$$

Then $\bar{S}_n(\theta) = F_*(Q, Q_n)$. By least squares, $Q_n \geq \hat{Q}_n$. By⁵ Lemma 4,

$$\bar{S}_n(\theta) = F_*(Q, Q_n) \geq F_*(Q, \hat{Q}_n) \geq F_*(\hat{Q}_n, \hat{Q}_n) = \bar{S}_n(\hat{\theta}_n).$$

Thus $\hat{\theta}_n$ is the MLE.

Let $0 < \epsilon < 1/2$, $f_\epsilon(\mu) = (1 - \epsilon)f(\mu) - \epsilon Q^*$, and $F_\epsilon(\theta) = F_\epsilon(\mu, Q) = F_*(Q, f_\epsilon(\mu))$. Since $f_\epsilon(\mu) \geq (1 - 2\epsilon)f(\mu)$,

$$F_\epsilon(\mu, (1 - 2\epsilon)Q) \geq \frac{1}{2} \log \det((1 - 2\epsilon)I) + F(\mu, Q),$$

hence $F_\epsilon(\theta)$ is proper. Let $\bar{X}_n = (X_1 + \dots + X_n)/n$ be the sample mean.

By the law of large numbers and (11) with $\mu = \mu^*$, there is $N \geq 1$, depending on the samples, such that, with probability one,

$$(\bar{X}_n - \mu^*) \otimes (\bar{X}_n - \mu^*) \leq \epsilon^2 Q^*, \quad \text{for } n \geq N,$$

and

$$\frac{1}{n} \sum_{k=1}^n (X_k - \mu^*) \otimes (X_k - \mu^*) \geq (1 - \epsilon)Q^*, \quad \text{for } n \geq N.$$

With $w = \bar{X}_n - \mu^*$ and $v = \mu - \mu^*$, using (9),

$$v \otimes w + w \otimes v \leq \epsilon(\mu^* - \mu) \otimes (\mu^* - \mu) + \frac{1}{\epsilon}(\bar{X}_n - \mu^*) \otimes (\bar{X}_n - \mu^*) \leq \epsilon f(\mu)$$

for all μ and $n \geq N$. Writing $X_k - \mu = (X_k - \mu^*) - v$,

$$\frac{1}{n} \sum_{k=1}^n (X_k - \mu) \otimes (X_k - \mu) - \frac{1}{n} \sum_{k=1}^n (X_k - \mu^*) \otimes (X_k - \mu^*)$$

equals $v \otimes v - (v \otimes w + w \otimes v)$. Combined with the last two inequalities, this leads to

$$\frac{1}{n} \sum_{k=1}^n (X_k - \mu) \otimes (X_k - \mu) \geq f_\epsilon(\mu)$$

for all μ and $n \geq N$. Since this implies

$$F_\epsilon(\theta) \leq \bar{S}_n(\theta), \quad \text{for all } \theta \text{ and } n \geq N,$$

this establishes A.

6. DISCLOSURES

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⁵Since $\det(Q^* + \delta I) \rightarrow \det Q^*$ as $\delta \rightarrow 0+$, $F_*(Q, Q^*) \geq F_*(Q^*, Q^*)$ is valid for $Q^* \geq 0$ by continuity.

A. PROOFS OF LEMMAS

Proof of Lemma 1. Assume $t \neq 0$. If v is orthogonal to an eigenvector x of Q , then we reduce to a lower-dimensional case by restricting to $x^\perp$. Hence we may assume $v \cdot x \neq 0$ for all eigenvectors x of Q . If v is orthogonal to an eigenvector x' of $Q' = Q + tv \otimes v$, then x' is an eigenvector of Q , contradicting our assumption. Hence we may assume $v \cdot x' \neq 0$ for all eigenvectors x' of Q' . If x'_1 and x'_2 are linearly independent eigenvectors of Q' corresponding to the same eigenvalue, then, for suitable r_1, r_2 , $x' = r_1 x'_1 + r_2 x'_2$ is an eigenvector of Q' satisfying $v \cdot x' = 0$, contradicting our assumptions. Hence we may assume the eigenvalues of Q' are simple. If x and x' are eigenvectors for Q and Q' corresponding to the same eigenvalue, then

$$t(v \cdot x')(v \cdot x) = x' \cdot (tv \otimes v)x = x' \cdot Q'x - x' \cdot Qx = 0,$$

contradicting our assumptions. Hence we may assume the eigenvalues of Q and the eigenvalues of Q' do not overlap. Let $\lambda_1, \lambda_2, \dots, \lambda_d$ be the eigenvalues of Q and $\lambda'_1, \lambda'_2, \dots, \lambda'_d$ the eigenvalues of Q' . Let $v_1, v_2, \dots, v_d$ be the corresponding orthonormal basis of eigenvectors of Q , and let⁶

$$p(z) = \det(Q - zI)(1 + tv \cdot (Q - zI)^{-1}v) = \prod_{i=1}^d (\lambda_i - z) \left(1 + t \sum_{i=1}^d \frac{(v \cdot v_i)^2}{\lambda_i - z} \right).$$

If λ' is an eigenvalue of Q' with eigenvector x' , then

$$\begin{aligned} (v \cdot x') (1 + t(v \cdot (Q - \lambda'I)^{-1}v)) &= v \cdot (x' + t(v \cdot x')(Q - \lambda'I)^{-1}v) \\ &= (Q - \lambda'I)^{-1}v \cdot ((Q - \lambda'I)x' + t(v \cdot x')v) \\ &= (Q - \lambda'I)^{-1}v \cdot (Q' - \lambda'I)x' = 0, \end{aligned}$$

hence $p(\lambda') = 0$. Since $\lambda'_1, \lambda'_2, \dots, \lambda'_d$ are simple, $p(z) = \prod_{i=1}^d (\lambda'_i - z)$. Since the product of the roots is $p(0)$, the result follows. $\square$

Proof of Lemma 2. Assume the result holds for Z . By Lemma 1,

$$\begin{aligned} \det(S(Z + v \otimes v)S) &= \det(SZS + Sv \otimes Sv) \\ &= \det(SZS)(1 + Sv \cdot (SZS)^{-1}Sv) \\ &= \det(Z) \det(S)^2 (1 + v \cdot Z^{-1}v) = \det(Z + v \otimes v) \det(S)^2, \end{aligned}$$

hence the result holds for $Z + v \otimes v$. If $Z = \delta I$, the result holds. Since every $Z > 0$ is a sum of δI and a sum of matrices of the form $v \otimes v$, the result follows by induction over the number of terms in the sum. $\square$

Proof of Lemma 3. The first part is checked by differentiation. For the second part, clearly $h(\lambda) \leq c$ implies $\lambda \leq 1/\epsilon$. Since $\lambda \log \lambda \geq -1/e$ and $(e-1)e^c \geq ec$ for $c \geq 1$, $h(\lambda) \leq c$ and $c \geq 1$ implies $\lambda \geq (1-1/e)/c \geq \epsilon$. $\square$

Proof of Lemma 4. Since $Q^* > 0$, there is $S > 0$ with $Q^* = S^2$. Let $Z = S^{-1}QS^{-1}$ and let $\lambda_1, \lambda_2, \dots, \lambda_d$ be the eigenvalues of Z . Then $\text{trace}(Q^{-1}Q^*) = \text{trace}(SQ^{-1}S) = \text{trace}(Z^{-1})$. By Lemmas 2 and 3,

$$\begin{aligned} 2F_*(Q, Q^*) - 2F_*(Q^*, Q^*) &= \log \det Q - \log \det Q^* + \text{trace}(Q^{-1}Q^* - I) \\ &= \log \det(SZS) - \log \det(S)^2 + \text{trace}(Z^{-1} - I) \\ &= \log \det Z + \text{trace}(Z^{-1} - I) \\ &= \sum_{i=1}^d (h(\lambda_i) - 1) \geq 0, \end{aligned} \tag{12}$$

establishing $F_*(Q, Q^*) \geq F_*(Q^*, Q^*)$. If (12) is an equality, $Z = I$ hence $Q = Q^*$. Now suppose $F_*(Q, Q^*) \leq c$. By (12), $h(\lambda_i) \leq \log(1/\epsilon)$ for $i = 1, 2, \dots, d$. By Lemma 3, $\epsilon \leq \lambda_i \leq 1/\epsilon$ for $i = 1, 2, \dots, d$. This implies $\epsilon I \leq Z \leq I/\epsilon$, hence $\epsilon Q^* \leq Q \leq Q^*/\epsilon$. $\square$

⁶The equation $1 + tv \cdot (Q - zI)^{-1}v = 0$ for z is the *secular equation*.

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